

NAME: _____

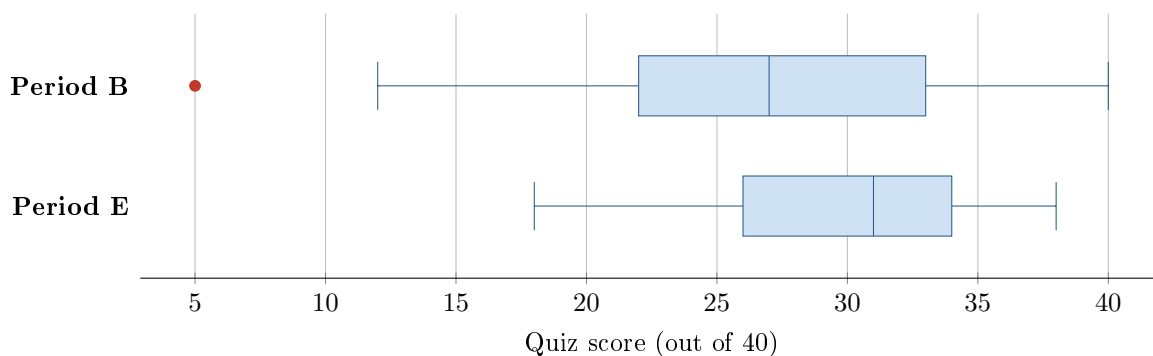
PERIOD: _____

Which Class Did Better?

AP Statistics · Unit 1 · Topic 1.9

[STAR] THE PROBLEM

Two AP Statistics classes took the same 40-point quiz. The side-by-side boxplots show the scores. Two students read them differently. **Sam:** “Period B did better — its highest score is 40 and Period E’s is only 38.” **Tia:** “Period E did better — its median is higher and its scores are more consistent.”



FIRST TAKE — one sentence before you work the parts

Which class did better on the quiz? One sentence, and name ONE feature of the plot you used.

[BOOK] COMPARING DISTRIBUTIONS (keep on the page)

Use comparative words such as higher, lower, more variable, or similar. Compare center (median), variability ($IQR = Q_3 - Q_1$, range = maximum minus minimum), apparent shape, and unusual features in context. Include plotted outliers when finding the overall range. A boxplot hides values within each quartile, so qualify shape claims with ‘appears.’ A median describes center; a maximum describes the highest observed value.

Work (a)–(c).

DOK 1 (a) Read the median for each class.

DOK 2 (b) In two or three sentences, compare the quiz scores' center, spread, apparent shape, and unusual features. Use points and comparative words.

DOK 3 (c) Whose claim is better supported, Sam's or Tia's? Cite two specific features of the boxplots that settle it. Then: if you could report only ONE statistic per class to a parent asking 'how did the class do?', which statistic would you choose, and why that one rather than the maximum?

Frame: _____'s claim is better supported because the _____ of Period E is _____ than Period B's, and Period E's scores are _____.

Frame: I would report the _____ because the maximum describes _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.

Optional #1 — not collected

DOK 2 Two bus routes were timed for 20 mornings each. Route 1: mean 24.0 min, SD 6.5 min, median 22 min. Route 2: mean 24.5 min, SD 1.8 min, median 24 min. Write two sentences comparing the routes for a student who must not be late, using comparative language and the statistics given.